

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Civil Engineering) SEMESTER - 2 Summer 2025 ( Regular )**

**Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 2**

**Subject Code & Name: 24AF1EGES204 - ENGINEERING GRAPHICS**

**Time : 4 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 A diagonal scale is mainly used for:  
A) Measuring curved surfaces B) Measuring straight lines only  
C) Measuring small subdivisions of a unit D) Enlarging drawings
- 2 What is the function of tolerances in design?  
A) Increase part weight B) Allow part interchangeability  
C) Reduce strength D) Eliminate manufacturing
- 3 In the aligned system of dimensioning, dimensions are placed:  
A) Always horizontal B) Always vertical  
C) Parallel to the dimension line D) On the border of the drawing
- 4 A point lies above the Horizontal Plane (HP) and in front of the Vertical Plane (VP). It is in which quadrant?  
A) First B) Second  
C) Third D) Fourth
- 5 If a line is perpendicular to HP and parallel to VP, its front view will be:  
A) Point B) True length  
C) Inclined D) Shorter than true length
- 6 When a line is inclined to HP but parallel to VP, its top view shows:  
A) True length B) Shortened length  
C) Point D) Horizontal line
- 7 When a circular plate lies on HP and is inclined to VP, its front view will appear as:  
A) Circle B) Line  
C) Ellipse D) Rectangle

8 The shape seen when a cylinder is resting on HP with its axis perpendicular to HP is:

- A) Circle in top view
- B) Rectangle in top view
- C) Ellipse in top view
- D) Triangle in top view

9 A prism is a solid bounded by:

- A) Only triangular faces
- B) Two parallel bases and rectangular lateral faces
- C) Curved surface only
- D) One base and curved surface

10 When a solid is cut by a plane, the shape of the section depends on:

- A) The position of the plane
- B) The material of the solid
- C) The color of the solid
- D) The surface texture of the solid

11 In first-angle projection, the object is placed:

- A) Between the observer and the plane
- B) Behind the projection plane
- C) In the third quadrant
- D) Above the projection plane

12 In isometric projection, the angle between any two isometric axes is:

- A)  $60^\circ$
- B)  $90^\circ$
- C)  $120^\circ$
- D)  $30^\circ$

Q2. Solve the following.

- A) Differentiate Aligned and Uni-directional system of placing the dimensions on a drawing with the help of diagrams. 6
- B) Construct a forward vernier scale of RF = 1/50 to read centimetres and long enough to measure 5 metres. 6

Q3. Solve the following.

- A) Draw the FV, TV and RHSV of the following points: (i) Point P lies in the HP and 22 mm behind the VP. (ii) Point Q lies in the VP and 32 mm below the HP. (iii) Point R lies 32 mm below the HP and 22 mm behind the VP. 6
- B) A line AB, 50 mm long, lies in the HP and makes an angle of  $45^\circ$  to the VP. Its end A is nearer to the VP and 25 mm in front of it. Draw the projections. 6

Q4. Solve Any One of the following.

- A) A regular pentagon ABCDE of side 30 mm has one of its edges parallel to the VP and inclined at  $30^\circ$  to the HP. The pentagon is inclined at  $45^\circ$  to the VP. Draw the projections. 12
- B) A pentagonal pyramid of edge of base 30 mm and length of axis 65 mm is resting on a corner of the base on the HP. The triangular face opposite to the corner on the HP is inclined to the HP at  $45^\circ$  with its shorter edge inclined to the VP at  $60^\circ$ . Draw its projections. 12

Q5. Solve Any One of the following.

- A) A triangular pyramid with a base side of 50 mm and a slant height of 70 mm, rests on the base on the HP with a side of base perpendicular to the VP. It is cut by an AIP inclined at  $30^\circ$  to the HP and bisecting the axis and a profile section plane intersecting the AIP at the edge of the pyramid parallel to the VP. Draw FV, sectional TV and sectional SV. 12

- B) A hexagonal pyramid of side of base 60 mm and length of axis 140 mm is kept on the ground on its base. It is cut by an AIP inclined at  $45^\circ$  to the base and cutting the axis at 94 mm from the apex. Draw the development of lateral surfaces of the pyramid. 12

Q6. Solve Any One of the following.

- A) Draw (a) FV in direction X, (b) TV and, (c) LHSV of the object as shown in Figure 1, using first angle method: 12

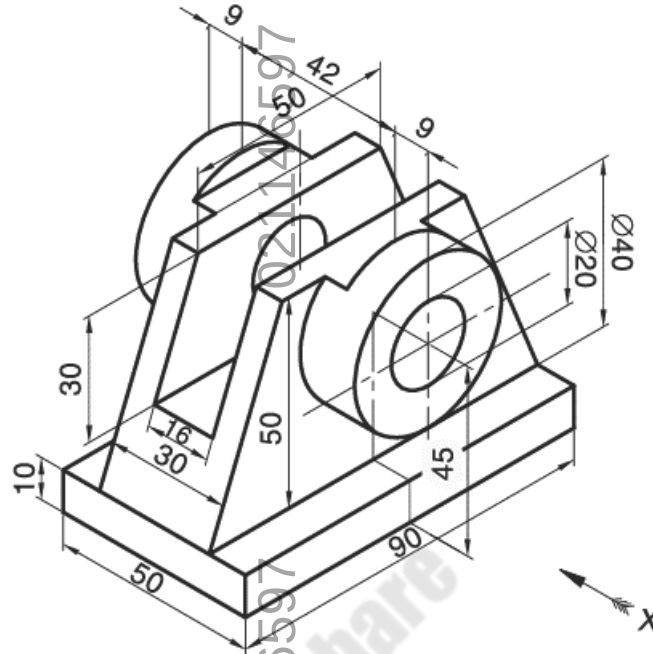


Figure 1

Draw the isometric view from the orthographic projections, as shown in Fig. 2 12

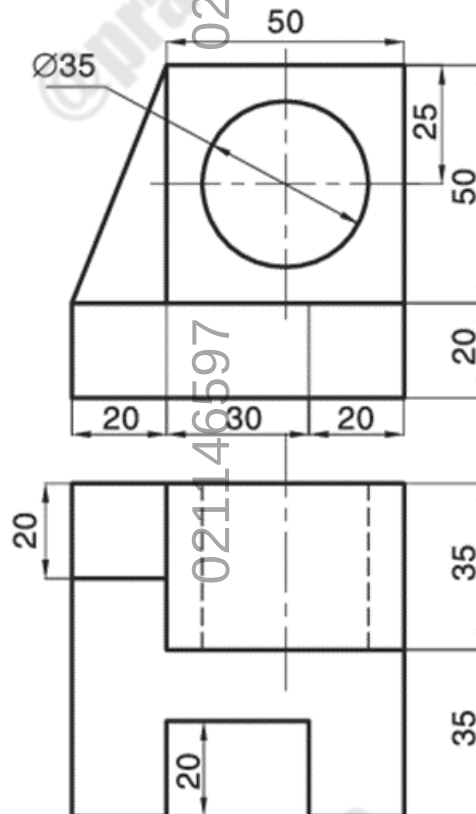


Figure 2

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